
Analog IC Design

Lecture 05 MOSFET Small Signal Model

Dr. Hesham A. Omran

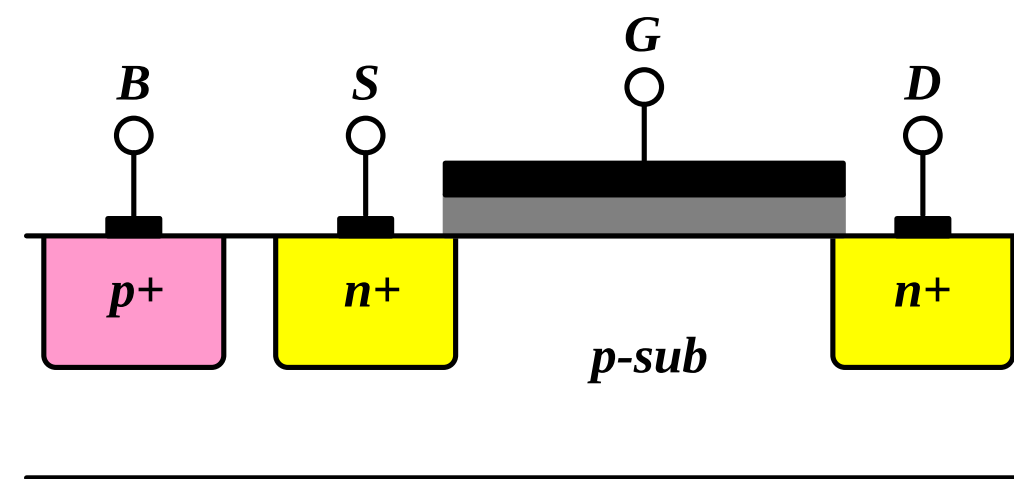
Integrated Circuits Lab (ICL)
Electronics and Communications Eng. Dept.
Faculty of Engineering
Ain Shams University

Outline

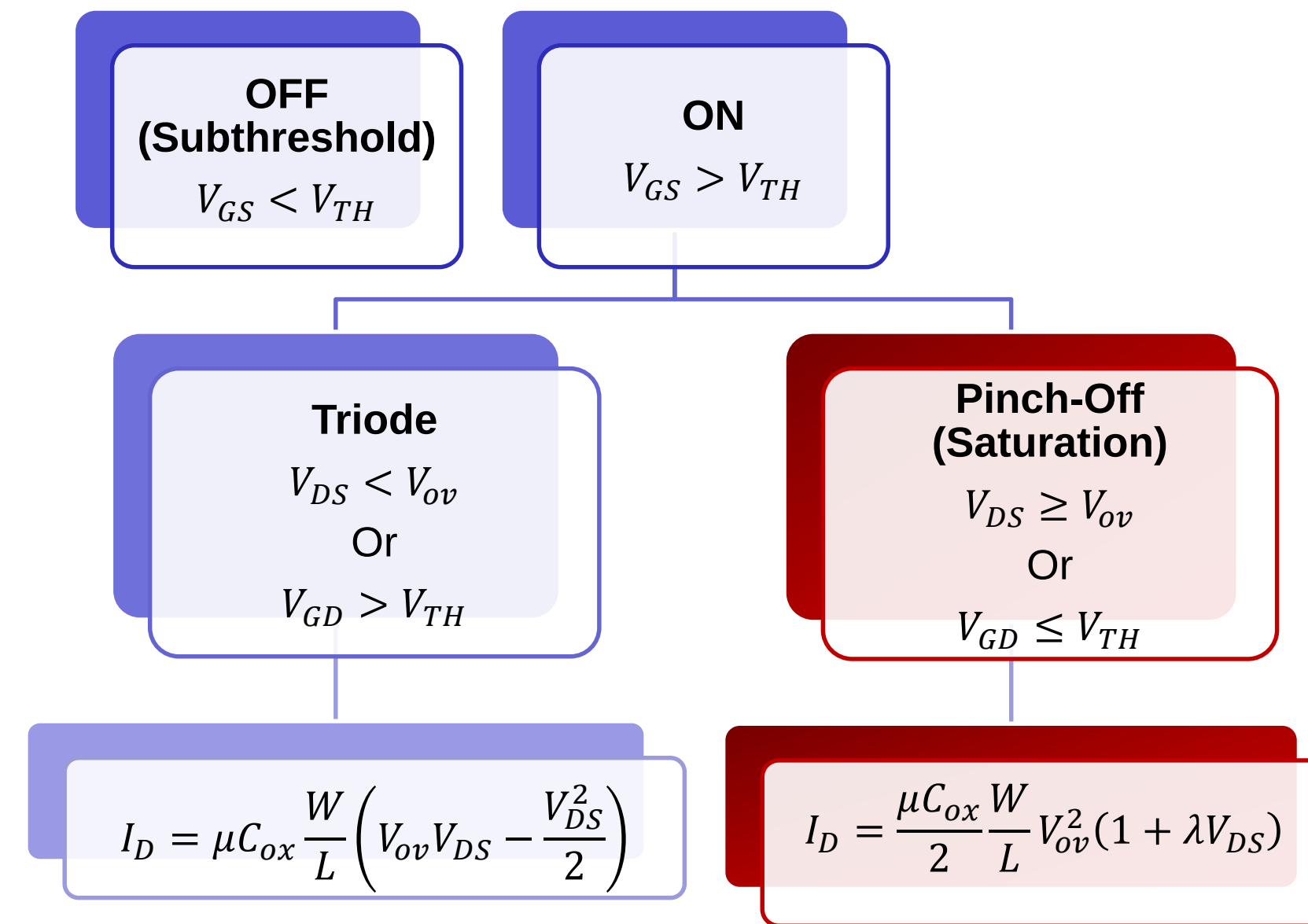
- ☐ Recapping previous key results
- ☐ The small signal approximation
- ☐ The transconductance (g_m)
- ☐ Body effect and body transconductance (g_{mb})
- ☐ Channel length modulation and output resistance ($r_o = \frac{1}{g_{ds}}$)
- ☐ Small signal model
- ☐ Short channel effects

N-Channel MOSFET Structure

- ❑ MOSFET: Metal-oxide-semiconductor field-effect transistor
- ❑ Three-terminal device: Gate (G), Source (S), and Drain (D)
- ❑ Substrate/Bulk/Body (S/B) can be treated as a fourth terminal



Regions of Operation Summary



MOSFET in Saturation

- The channel is pinched off if the difference between the gate and drain voltages is not sufficient to create an inversion layer

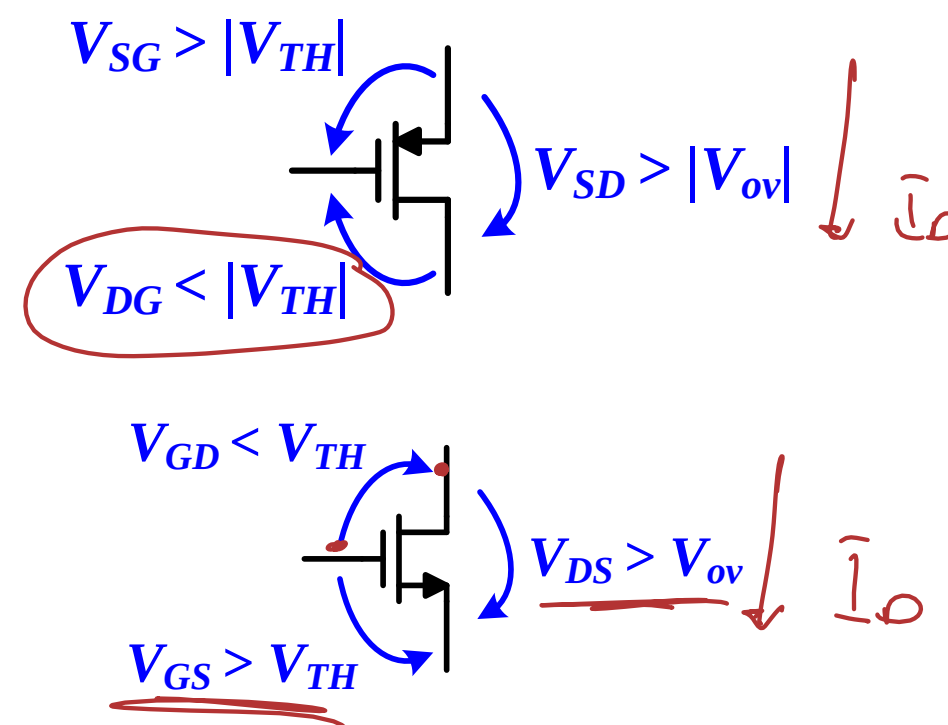
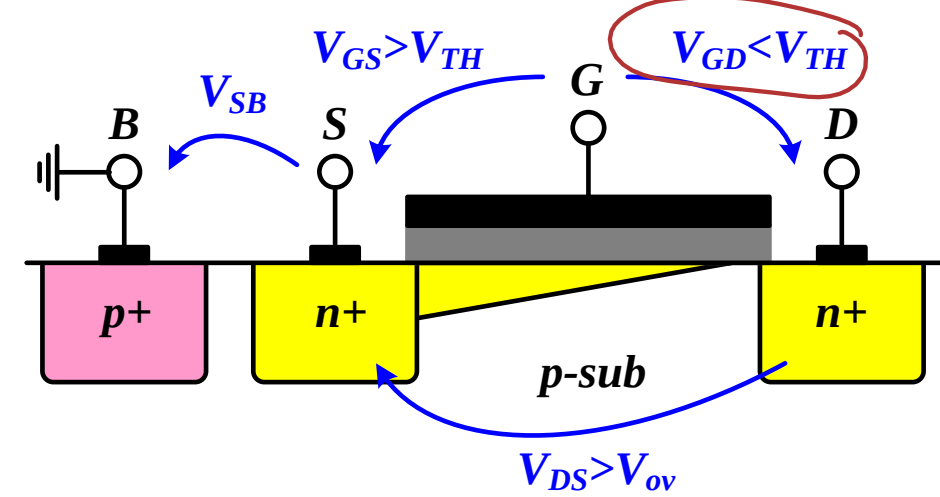
$$V_{GD} \leq V_{TH} \text{ OR } V_{DS} \geq V_{ov}$$

- Square-law (long channel MOS)

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{ov}^2 (1 + \lambda V_{DS})$$

$$V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$$

$\neq 0$



Large Signal Model

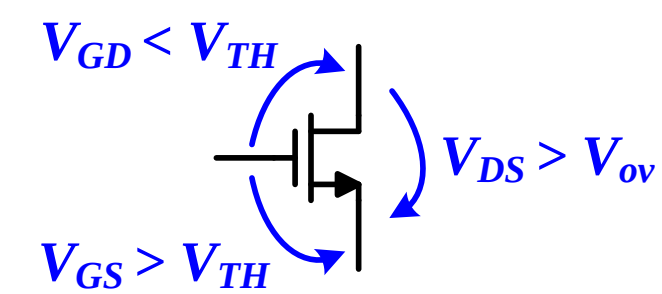
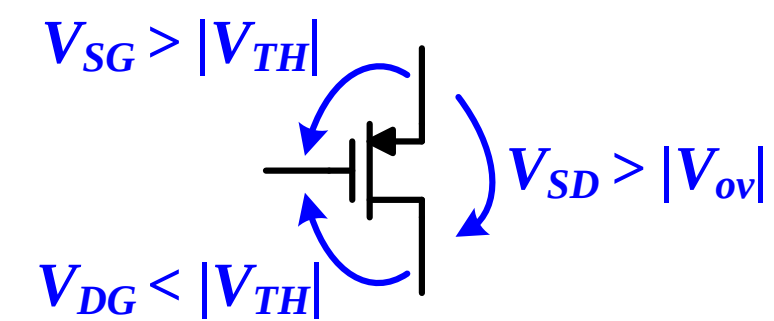
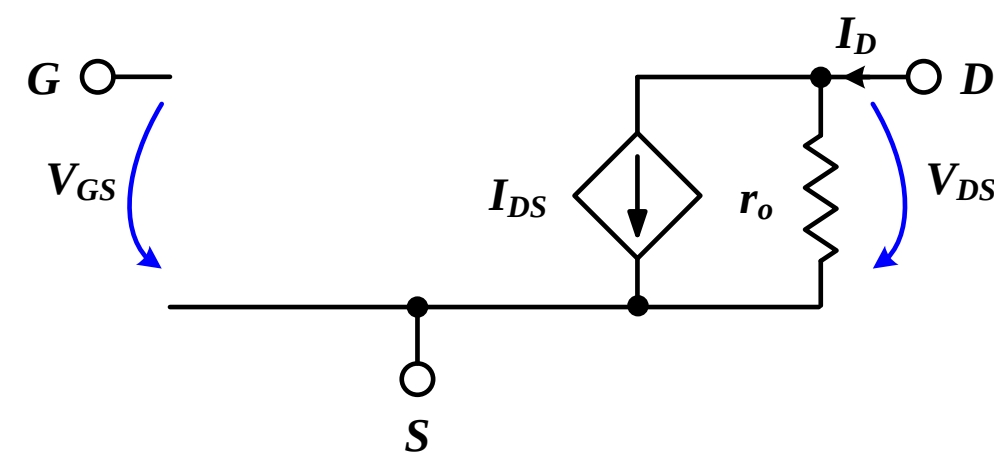
- ❑ The channel is pinched off if the difference between the gate and drain voltages is not sufficient to create an inversion layer

$$V_{GD} \leq V_{TH} \text{ OR } V_{DS} \geq V_{ov}$$

- ❑ Square-law (long channel MOS)

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{ov}^2 (1 + \lambda V_{DS})$$

$$V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$$



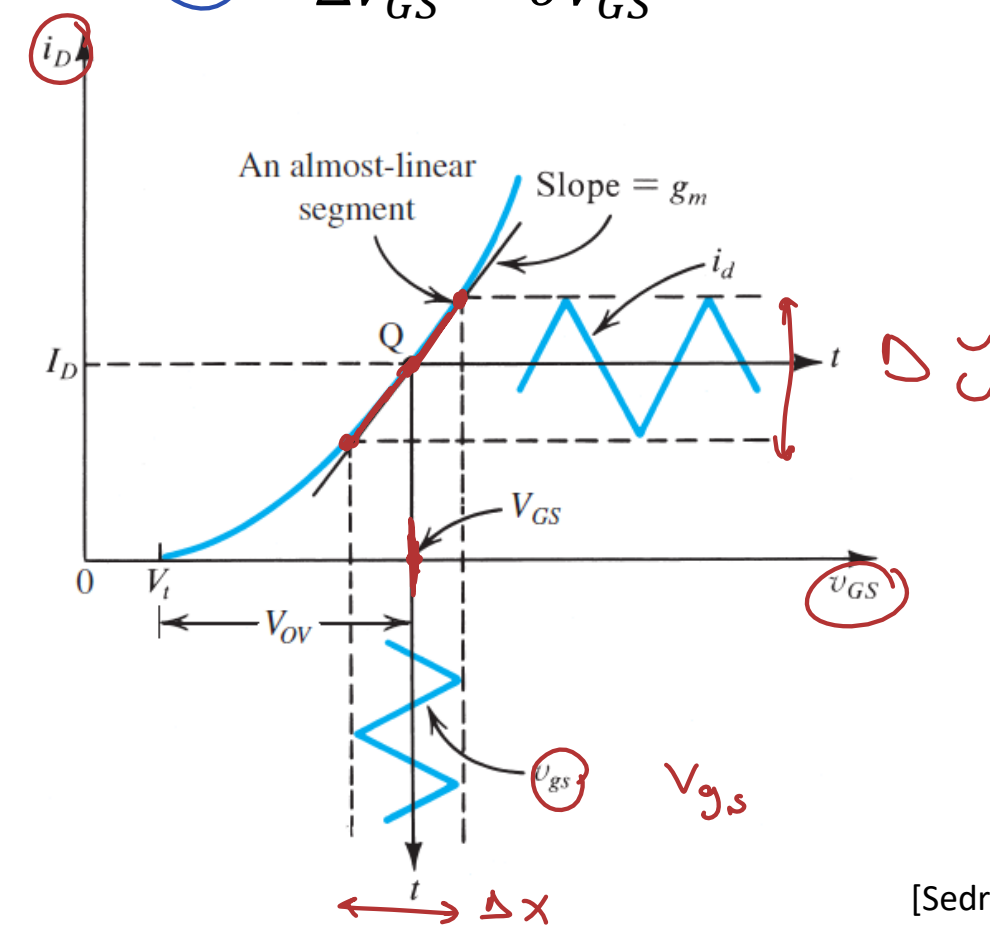
Small Signal Approximation

- ❑ The transistor is a VCCS
- ❑ Transconductance: how well it converts the voltage to a current

$$\frac{\Delta y}{\Delta x} = \text{slope}$$

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{\partial I_D}{\partial V_{GS}}$$

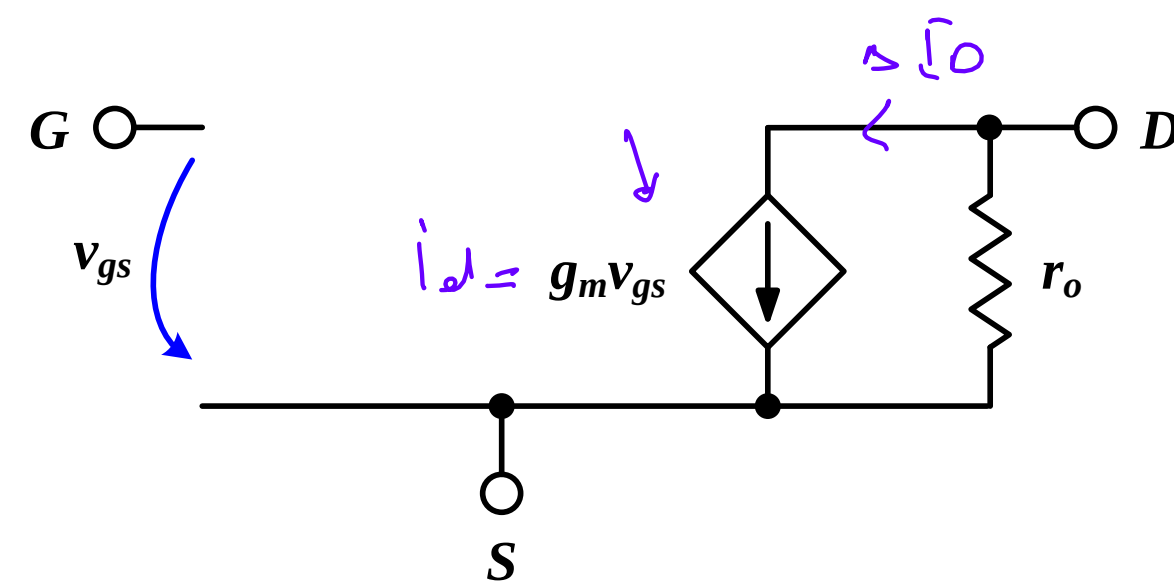
Q : Quiescent PT
 : OP PT
 : Bias PT



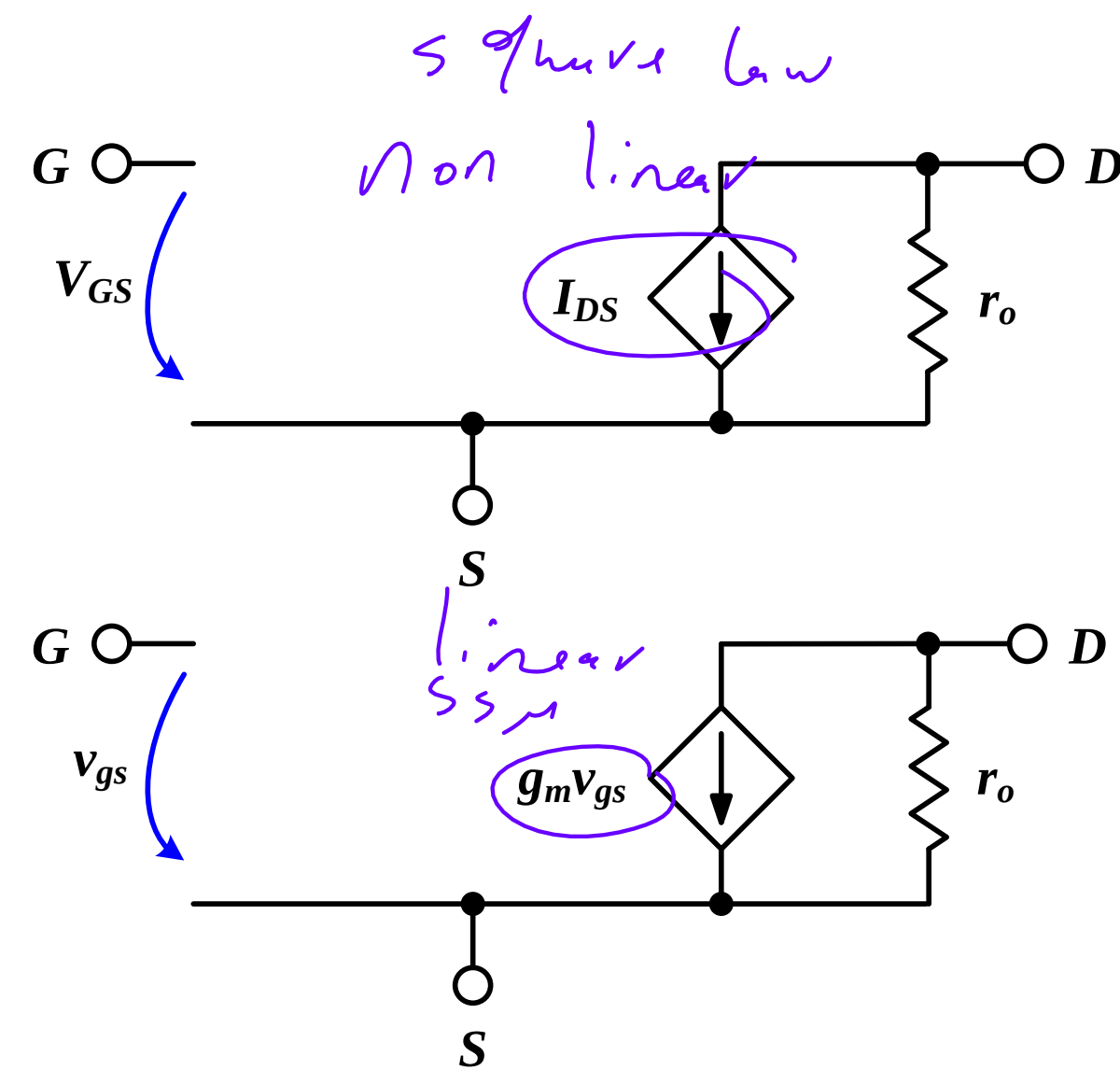
Small Signal Model

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{\partial I_D}{\partial V_{GS}}$$

$$r_o = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{1}{\frac{\partial I_D}{\partial V_{DS}}}$$



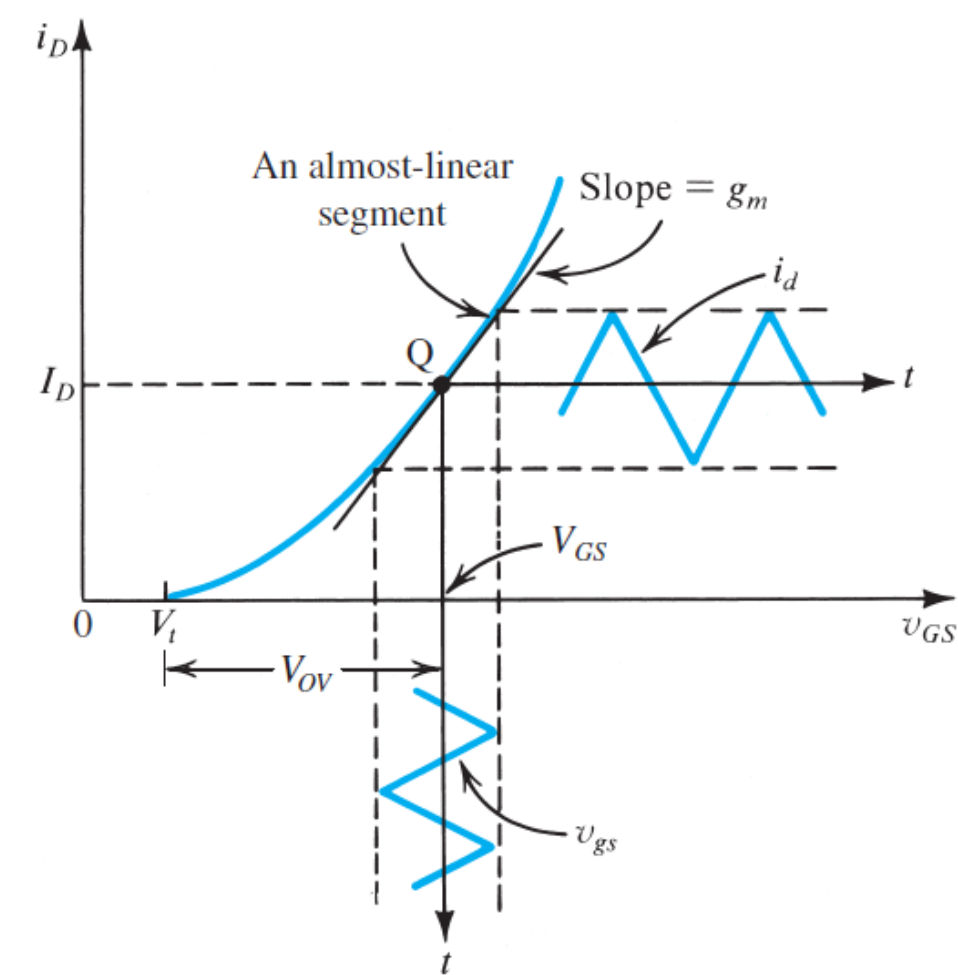
Large Signal vs Small Signal Model



Transconductance

- ❑ The transistor is a VCCS
- ❑ Transconductance: how well it converts the voltage to a current

$$\begin{aligned}
 I_D &\approx \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{ov}^2 \\
 g_m &= \frac{\Delta I_D}{\Delta V_{GS}} = \frac{\partial I_D}{\partial V_{GS}} = \frac{\partial I_D}{\partial V_{ov}} \\
 &= \mu C_{ox} \frac{W}{L} V_{ov} \\
 &= \sqrt{\mu C_{ox} \frac{W}{L} \cdot 2I_D} \\
 &= \frac{2I_D}{V_{ov}}
 \end{aligned}$$



• $g_m = K_n \quad V_{ov} = \frac{2 I_D}{g_m} = \sqrt{2 K_n I_D}$

Transconductance

$$I_D \approx \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{ov}^2$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox} \frac{W}{L} V_{ov} = \sqrt{\mu_n C_{ox} \frac{W}{L} \cdot 2I_D} = \frac{2I_D}{V_{ov}}$$

* لو معایا ۱ device
و بقیه من اار table

محکوم ب صلا معینه

<u>W/L constant</u>	<u>V_{ov} constant</u>	<u>I_D constant</u>
$g_m \propto V_{ov}$	$g_m \propto W/L$	$g_m \propto \sqrt{W/L}$
$g_m \propto \sqrt{I_D}$	$g_m \propto I_D$	$g_m \propto 1/V_{ov}$

Measurement

دول ۱ design

Body Effect

□ V_{SB} affects the charge required to invert the channel

- Increasing V_S or decreasing V_B increases V_{TH}

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\Phi_F + V_{SB}} - \sqrt{|2\Phi_F|} \right)$$

- Φ_F = surface potential at threshold

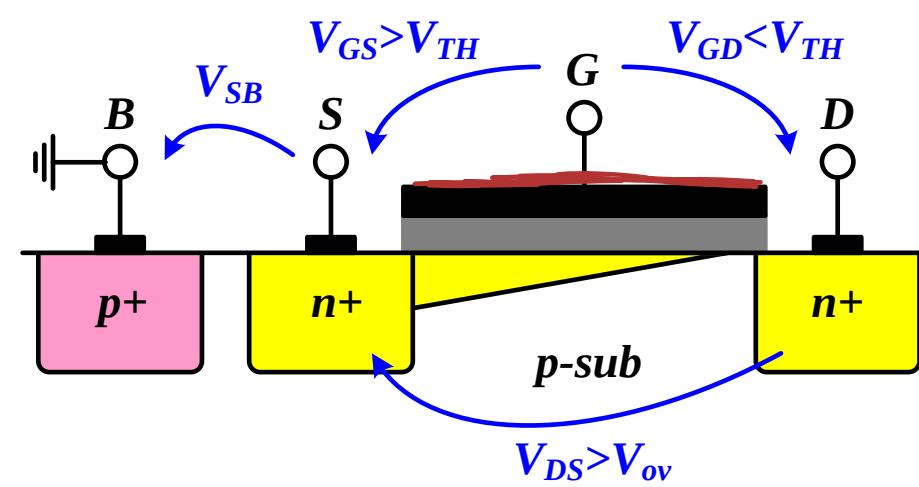
- Depends on doping level and intrinsic carrier concentration n_i

- γ = body effect coefficient

- Depends on C_{ox} and doping

$\sqrt{V_{SB}} \uparrow$ $\sqrt{V_{TH}} \uparrow$
 $\rightarrow \downarrow I_D$

$\Delta V_{GS} \xrightarrow{g_m} \Delta I_D$
 $\Delta V_{SB} \xrightarrow{g_{mb}} \Delta I_D$



Bulk Transconductance

- The bulk behaves as a second gate that changes the output current

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = \eta g_m$$

η is typically 0.1 $\rightarrow$ 0.25

کامتا انکسٹر فر ولو ہا؟ کا یکونہ کنری V_{BS} اہل
eff

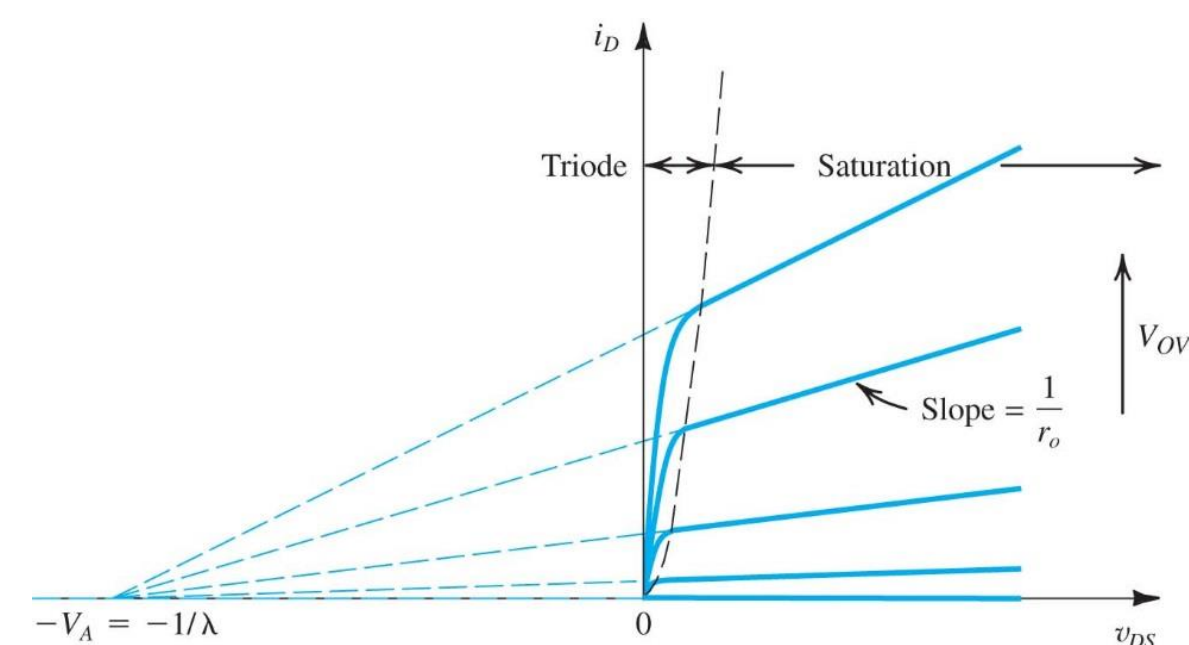
Channel Length Modulation (CLM)

- The VCCS is not ideal: There is some dependence on V_{DS}

$$r_o = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{1}{\partial I_D / \partial V_{DS}} = \frac{1}{g_{ds}} = \frac{V_A}{I_{DS}} = \frac{1}{\lambda I_{DS}}$$

V_A : Early voltage ($V_A \propto L$) $\leftrightarrow$ λ : Channel length modulation coefficient ($\lambda \propto 1/L$)

$$I_D = I_{DS} + \frac{V_{DS}}{r_o} = I_{DS} \left(1 + \frac{V_{DS}/I_{DS}}{r_o} \right) = \frac{\mu C_{ox} W}{2 L} V_{ov}^2 (1 + \lambda V_{DS})$$



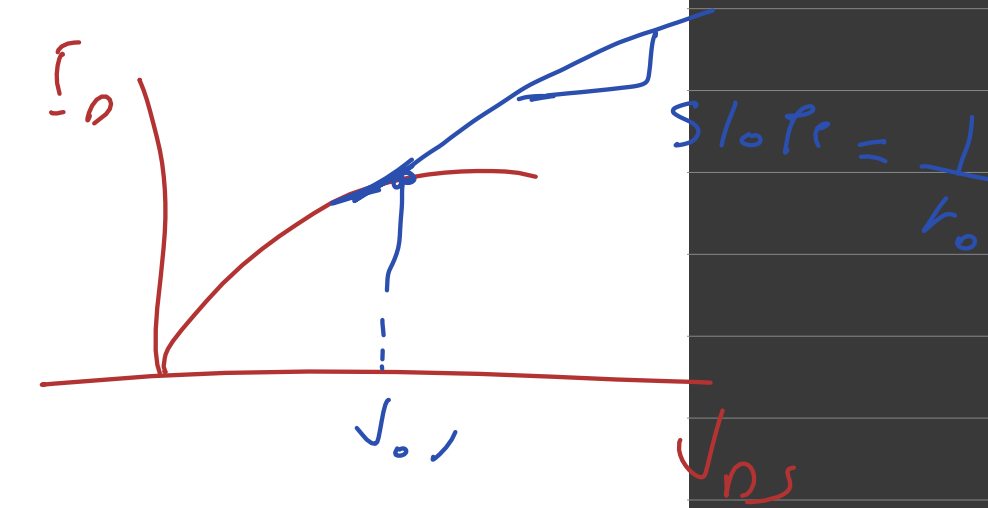
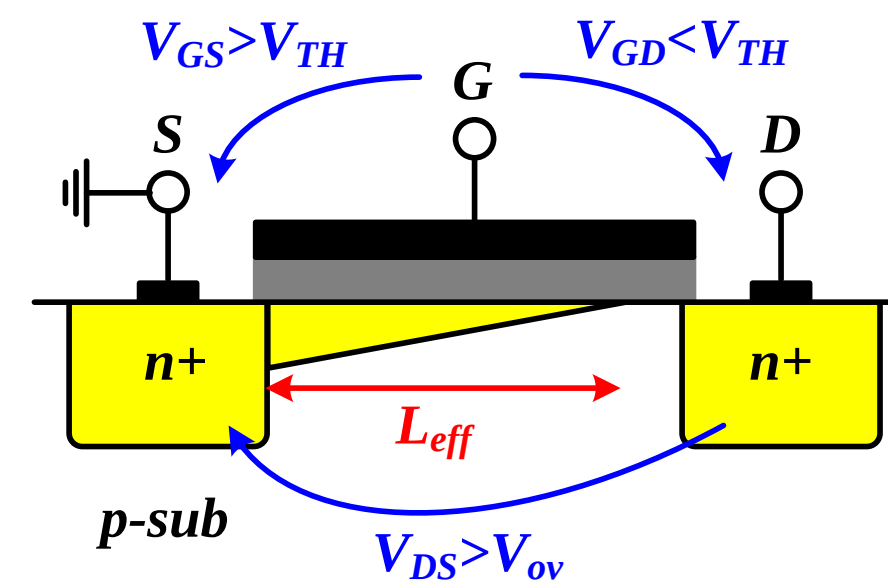
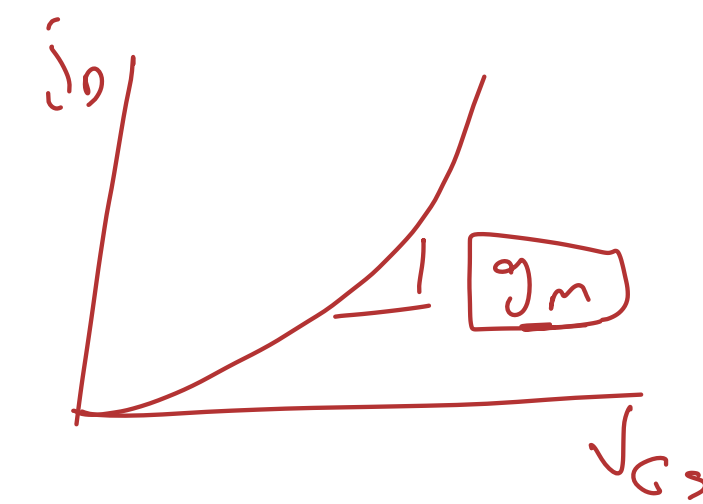
Channel Length Modulation (CLM)

- ❑ L_{eff} decreases with $V_{DS} \rightarrow$ Shorter L gives more current
- ❑ V_A : Early voltage ($V_A \propto L$)
- ❑ λ : Channel length modulation coefficient ($\lambda \propto 1/L$)

$$I_D = \frac{\mu C_{ox} W}{2 L} V_{ov}^2 (1 + \lambda V_{DS})$$

$$r_o = \frac{V_A}{I_{DS}} = \frac{1}{\lambda I_{DS}}$$

- ❑ V_A increases with V_{DS} : higher r_o as we go deeper into saturation



Handwritten notes in Persian:
 r_o منتهی به V_{DS} می‌رسد
 $r_o = f(V_{DS})$
 هر چه V_{DS} بیشتر شود، r_o بیشتر می‌شود

Handwritten note: $\text{slope} = \frac{1}{r_o}$

Low-Frequency Small-Signal Model

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu C_{ox} \frac{W}{L} V_{ov} = \sqrt{\mu C_{ox} \frac{W}{L} \cdot 2I_D} = \frac{2I_D}{V_{ov}}$$

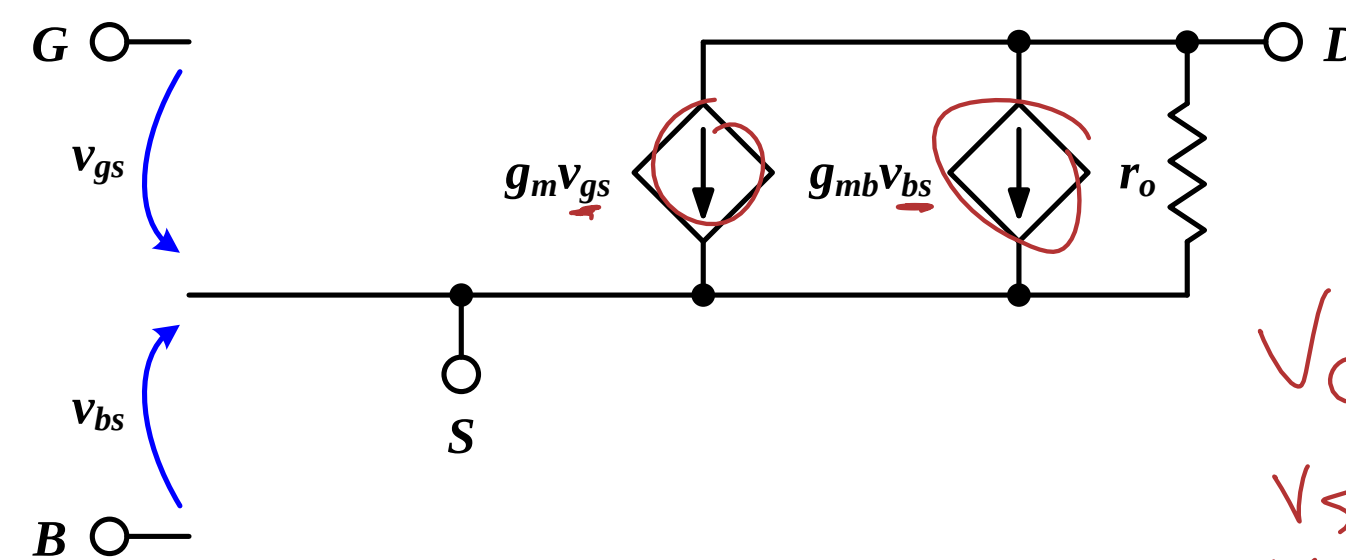
$$g_{mb} = \eta g_m$$

$$\eta \approx 0.1 - 0.25$$

$$r_o = \frac{1}{\partial I_D / \partial V_{DS}} = \frac{V_A}{I_D} = \frac{1}{\lambda I_D}$$

$$V_A \propto L \leftrightarrow \lambda \propto \frac{1}{L}$$

$$V_{DS} \uparrow \rightarrow V_A \uparrow \rightarrow r_o \uparrow$$



$V_{GS} \uparrow \rightarrow I_D \uparrow$
 $V_{SB} \uparrow \rightarrow I_D \downarrow$
 $V_{BS} \uparrow \rightarrow I_D \uparrow$

* V_{SB}

* V_{BS}

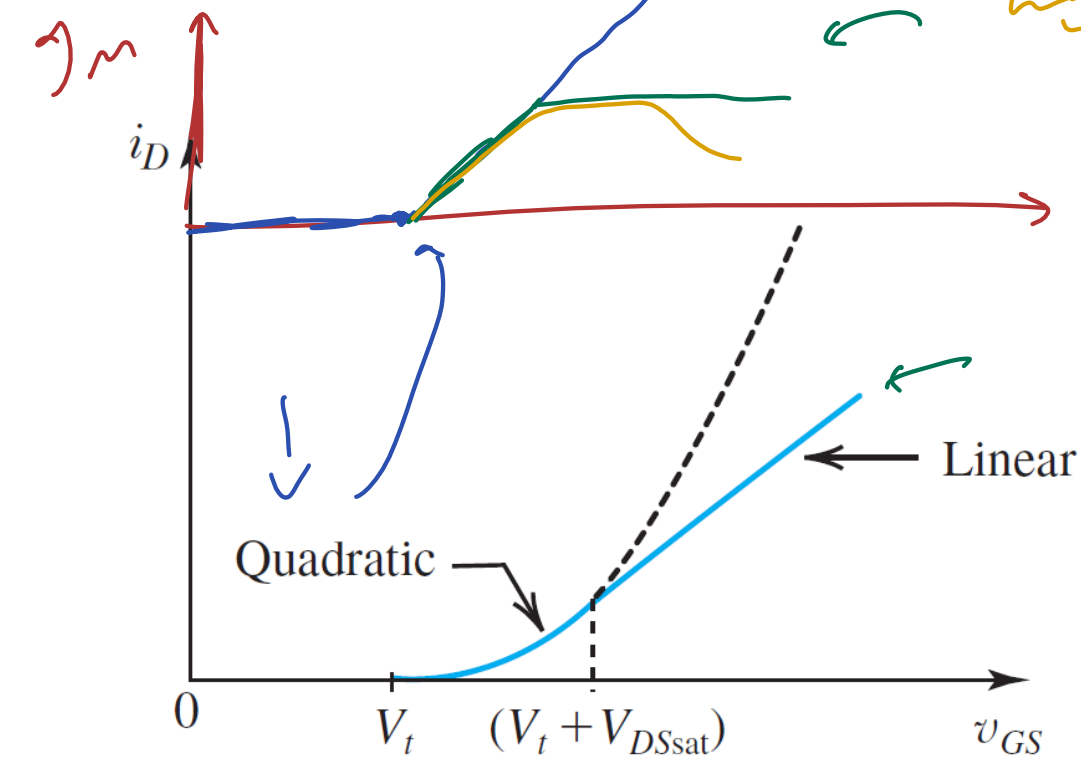
Large Signal

Small Signal

Short Channel Effects: Velocity Saturation

□ ID-VGS quadratic: $g_m = \frac{\partial I_D}{\partial V_{GS}} = \text{linear} \rightarrow g_m$ increases with V_{GS}

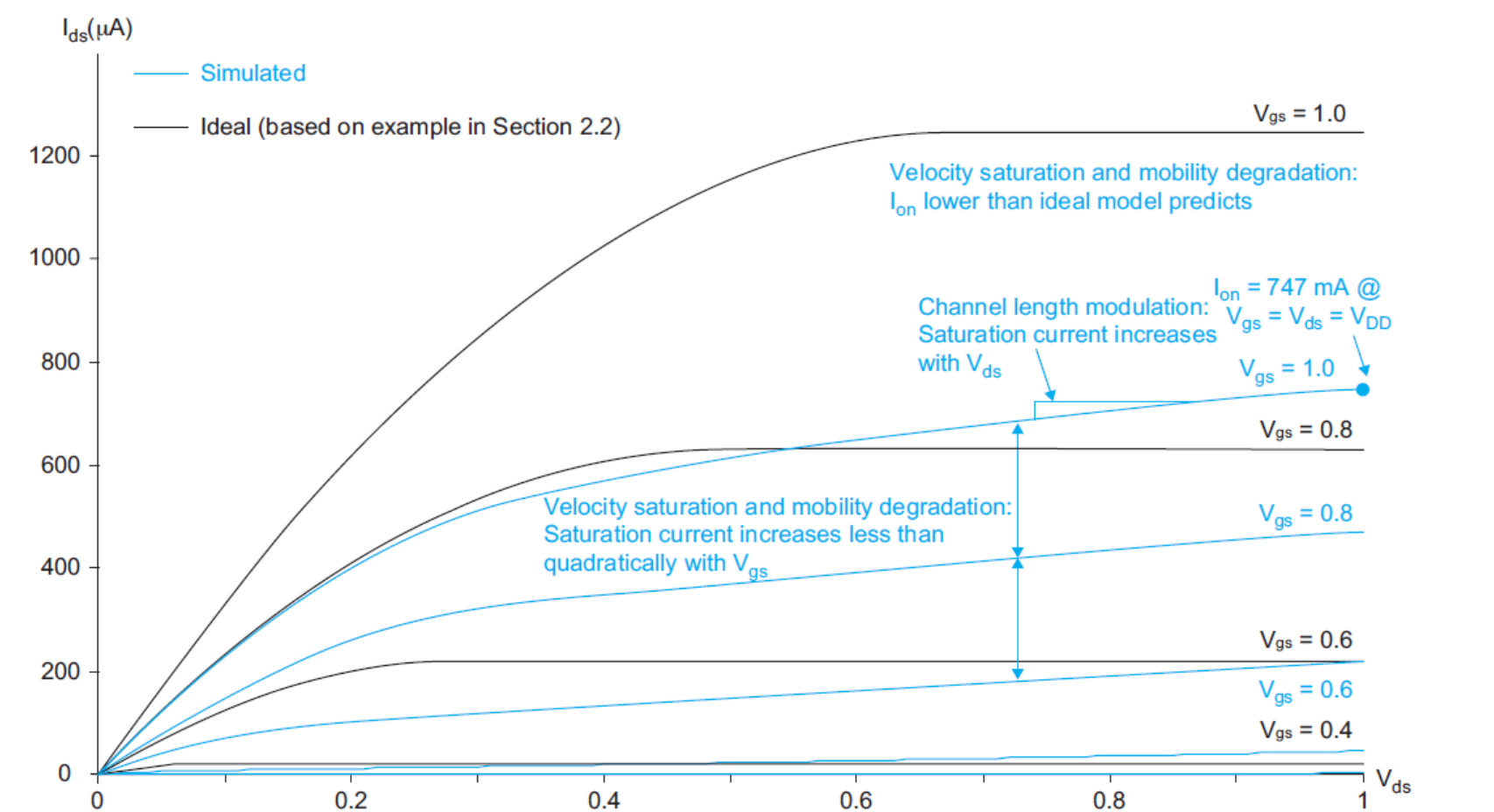
□ ID-VGS linear: $g_m = \frac{\partial I_D}{\partial V_{GS}} = \text{constant} \rightarrow g_m$ saturates
velocity sat



بعد از این نقطه g_m ثابت میماند
 به دلیل کاهش قابلیت
 هدایت الکتریکی

Short Channel Effects: CLM and DIBL

- ID-VDS horizontal: no V_{DS} dependence $\rightarrow$ ideal current source
- More slope $\rightarrow$ more V_{DS} dependence $\rightarrow$ smaller $r_o \rightarrow$ smaller gain



for higher o/p resistance we need to use long channel devices, why?

BC 5 gain depends on $|r_o|$

when $r_o \downarrow$ $A_v \downarrow$

when $r_o \uparrow$ $A_v \uparrow$

Thank you!

References

- ❑ A. Sedra and K. Smith, “Microelectronic Circuits,” Oxford University Press, 7th ed., 2015.
- ❑ B. Razavi, “Fundamentals of Microelectronics,” Wiley, 2nd ed., 2014.
- ❑ B. Razavi, “Design of Analog CMOS Integrated Circuits,” McGraw-Hill, 2nd ed., 2017.
- ❑ N. Weste and D. Harris, “CMOS VLSI Design,” Pearson, 4th ed., 2010.